

Backend Project Handbook

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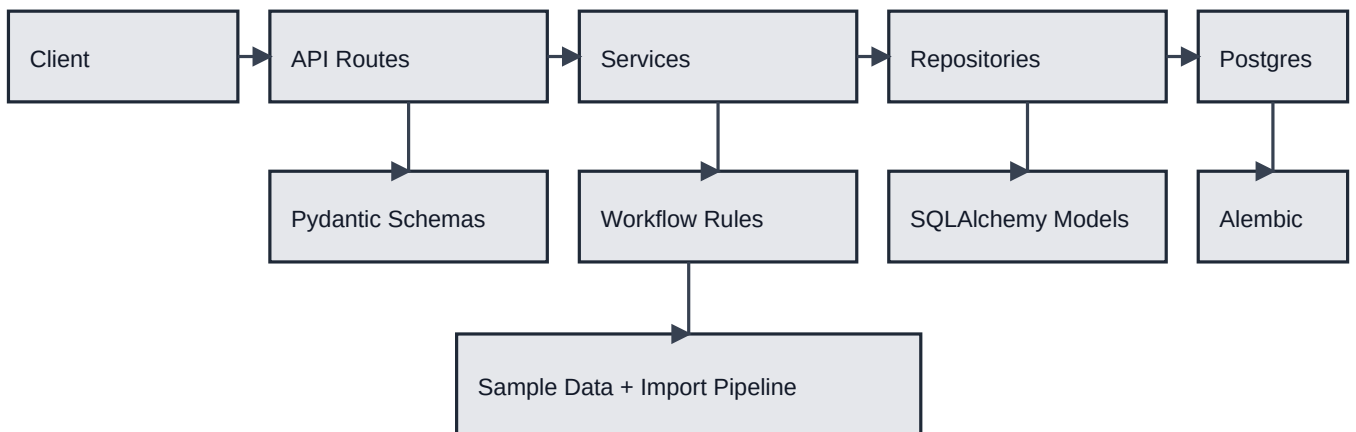
1. Executive Summary

This backend is a lead-centric sales intelligence system built on FastAPI, SQLAlchemy, Alembic, and PostgreSQL.

The lead is the central business entity and is connected to customer, campaign, journey, engagement, call, task, follow-up, AI prediction, and decision modules.

2. Architecture

System Architecture



Flow: request in -> domain orchestration -> persistent state -> typed response

- API Layer: app/api/routes
- Service Layer: app/services
- Repository Layer: app/repositories
- Persistence Layer: app/models + PostgreSQL
- Schema Contracts: app/schemas
- DB Evolution: Alembic migrations

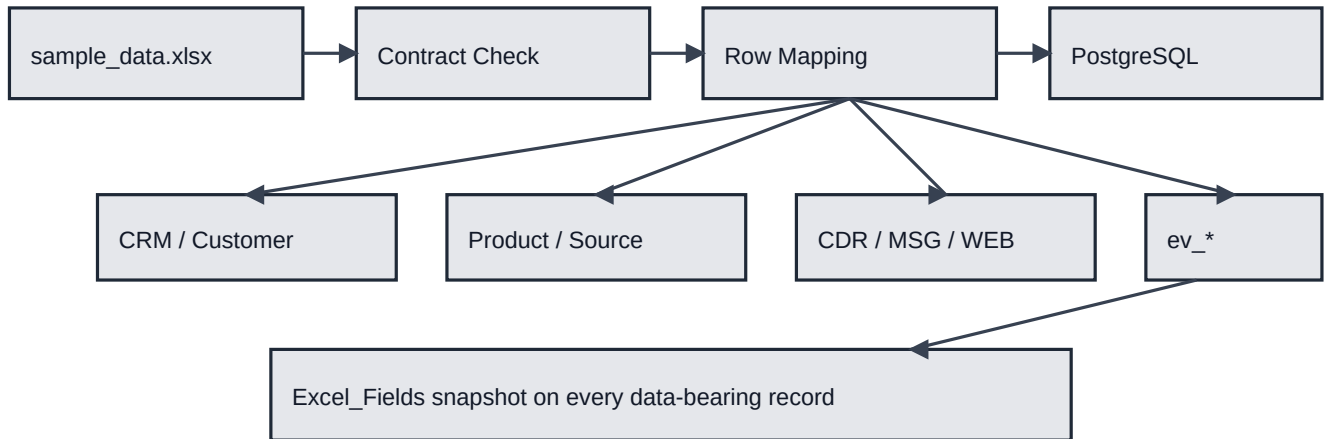
Request flow:

Client -> Route -> Service -> Repository -> DB
-> business rules -> response schema

Architecture Diagram (Mermaid):

Excel Ingestion Diagram:

Excel Ingestion And Persistence



3. Feature Modules

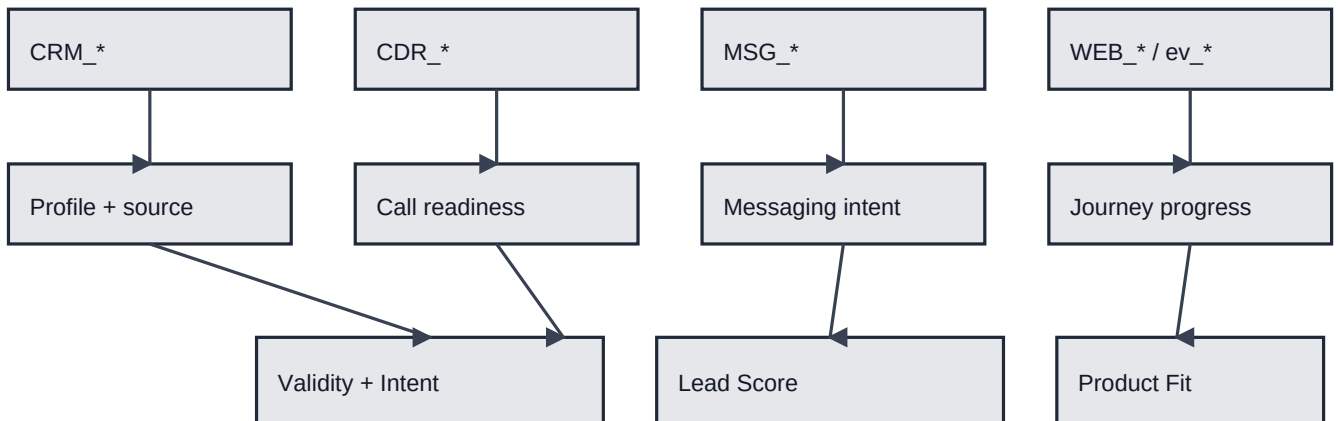
1. Health + Dashboard
2. Customers
3. Leads
4. Products
5. Campaigns
6. Engagement Channels
7. Website Journey
8. Calls
9. Tasks + Follow-ups
10. Data Import + Export
11. ML Training + Model Catalog
12. AI Predictions
13. Decision Engine
14. Analytics
15. Reports
16. Lead Lifecycle (assignment, transfer, outcomes, timeline, details)

4. Data Processing

1. Source data is loaded from backend/sample_data.xlsx using scripts/preload_sample_data.py.
2. The loader maps grouped records into products, campaigns, customers, leads, tasks, follow-ups, calls, engagement events, and website events.
3. Live APIs use these persisted records, not synthetic smoke values.
4. AI endpoints score/segment leads and store prediction logs.
5. Decision endpoint combines predictions + rules into next action recommendations.
6. Lifecycle module adds assignment/transfer, outcome tracking, and timeline events to create a connected business workflow.
7. Unified lead details endpoint aggregates lead + customer + campaign + engagement + AI + operational activity for frontend rendering.

Feature Mapping Diagram:

Excel Feature Groups To Business Features



Only exact source columns or deterministic derived outputs are allowed.

4.1 Sample Data Columns (sample_data.xlsx)

Total columns in dataset sheet: 91

1. Customer_ID
2. CRM_Channel
3. CRM_Department
4. CRM_Lead_Type
5. CRM_Lead_Source
6. CRM_Data_Medium
7. CRM_Source
8. CRM_Data_Source_Platform
9. CRM_Product_Code
10. CRM_Product_Name
11. CRM_UTM_Source
12. CRM_UTM_Medium
13. CRM_UTM_Campaign
14. CRM_Gender
15. CRM_Age_Band
16. CRM_Income_Band
17. CRM_Occupation
18. CRM_Education
19. CRM_Tobacco_User
20. CRM_NonResident_Flag
21. CRM_Existing_Plan_Flag
22. CRM_Lead_Create_Hour
23. CRM_Lead_Create_DayOfWeek

24. CRM_Lead_Create_Weekend
25. CRM_Lead_Create_Month
26. CRM_Days_Create_To_Update
27. CRM_Contact_Count
28. CRM_NonContact_Count
29. CRM_No_Of_Attempts
30. CRM_Has_Application_No
31. CRM_Has_Payment_Flag
32. CDR_Total_Calls
33. CDR_Connected_Calls
34. CDR_Connect_Rate
35. CDR_Total_Talk_Sec
36. CDR_Avg_Talk_Sec
37. CDR_Max_Talk_Sec
38. CDR_Distinct_Call_Days
39. CDR_Distinct_Agents
40. CDR_Callbacks_Scheduled
41. CDR_Inbound_Calls
42. CDR_Top_Call_Status
43. CDR_Top_Q_Type
44. CDR_First_Call_Delay_Days
45. CDR_Call_Span_Days
46. MSG_Campaigns_Targeted
47. MSG_Sent
48. MSG_Delivered
49. MSG_Read
50. MSG_Clicked
51. MSG_Replied
52. MSG_Failed
53. MSG_Engaged
54. WEB_Tracked
55. Visits
56. Page_Views
57. Total_Seconds_Spent
58. ev_Lead_Creation
59. ev_Lead_Submitted
60. ev_Calculator_Start
61. ev_Retrieve_Quote
62. ev_Quote_Proceed
63. ev_Personal_Details_Next
64. ev_Plan_Details_Load
65. ev_Make_Payment_Load
66. ev_Pay_Now
67. ev_Payment_Success

68. ev_Payment_Failure
69. ev_Proposal_Form_Load
70. ev_Proposal_Form_Proceed
71. ev_Generate_OTP
72. ev_OTP_Submit
73. ev_Brochure_Download
74. ev_Connect_With_Expert
75. ev_Page_Scroll_25
76. ev_Page_Scroll_95
77. WEB_Step_Name
78. WEB_Step_Number
79. WEB_New_Vs_Repeat
80. WEB_Plan_Type
81. WEB_Plan_Variant
82. WEB_Quoted_Price
83. WEB_Coverage_Amount
84. WEB_Payment_Frequency
85. WEB_Last_Touch_Channel
86. WEB_UTM_Source
87. WEB_Device_Type
88. Label_Source_Disposition
89. Label_Source_Lead_Status
90. Label_Basis
91. LABEL_Customer_Validity

4.2 Column Meaning (data_dictionary sheet)

- Customer_ID | source=generated | Random anonymized row id (mapping kept outside the dataset)
- CRM_Channel | source=CRM CHANNEL | Acquisition channel of the lead
- CRM_Department | source=CRM DEPARTMENT | Owning department
- CRM_Lead_Type | source=CRM LEAD_TYPE | Lead type code
- CRM_Lead_Source | source=CRM LEAD_SOURCE | Source system/partner of the lead
- CRM_Data_Medium | source=CRM DATA_MEDIUM | Data medium code
- CRM_Source | source=CRM SOURCE | Combined source string
- CRM_Data_Source_Platform | source=CRM P_DATA_SOURCE | Platform that generated the lead (e.g. PARTNER_APP)
- CRM_Product_Code | source=CRM PROD_ID | Product id attached to the lead
- CRM_Product_Name | source=CRM PRODUCT_NAME | Product name if recorded
- CRM_UTM_Source | source=CRM MISC_UTM_SOURCE | Marketing UTM source
- CRM_UTM_Medium | source=CRM MISC_UTM_MEDIUM | Marketing UTM medium
- CRM_UTM_Campaign | source=CRM MISC_UTM_CAMPAIGN | Marketing UTM campaign
- CRM_Gender | source=CRM CUST_GENDER | Declared gender (demographic, non-identifying)
- CRM_Age_Band | source=CRM CUST_AGE (banded) | Age banded: 18-25/26-35/36-45/46-55/56+/Unknown
- CRM_Income_Band | source=CRM ANNUAL_INCOME (banded) | Annual income banded: <3L/3-5L/5-10L/10-25L/25L+/Unknown

- CRM_Occupation | source=CRM OCCUPATION | Declared occupation
- CRM_Education | source=CRM EDUCATION | Declared education
- CRM_Tobacco_User | source=CRM TOBACCO_USER | Tobacco user flag Y/N
- CRM_NonResident_Flag | source=CRM NonResident_FLAG | NonResident flag
- CRM_Existing_Plan_Flag | source=CRM EXISTING_PLAN_PRODUCT | Existing plan product if any
- CRM_Lead_Create_Hour | source=CRM LEAD_CREATE_DATE (derived) | Hour of day the lead was created (0-23)
- CRM_Lead_Create_DayOfWeek | source=CRM LEAD_CREATE_DATE (derived) | Day of week of lead creation
- CRM_Lead_Create_Weekend | source=CRM LEAD_CREATE_DATE (derived) | 1 if created on Sat/Sun
- CRM_Lead_Create_Month | source=CRM LEAD_CREATE_DATE (derived) | Creation month YYYY-MM
- CRM_Days_Create_To_Update | source=CRM dates (derived) | Days between lead creation and last update
- CRM_Contact_Count | source=CRM CNT_COUNT | CRM-recorded contact count
- CRM_NonContact_Count | source=CRM NCNT_COUNT | CRM-recorded non-contact count
- CRM_No_Of_Attempts | source=CRM NO_OF_ATTEMPTS | CRM-recorded attempt count
- CRM_Has_Application_No | source=CRM APPLICATION_NO (flag) | 1 if an application number exists (id itself dropped)
- CRM_Has_Payment_Flag | source=CRM COLL_AMOUNT (flag) | 1 if payment was collected (amount dropped)
- CDR_Total_Calls | source=CDR Apr-Jun (agg) | Total dialer call records for this lead
- CDR_Connected_Calls | source=CDR (agg) | Calls with duration > 0 (a recording exists for these)
- CDR_Connect_Rate | source=CDR (agg) | Connected / total calls
- CDR_Total_Talk_Sec | source=CDR (agg) | Sum of call durations in seconds
- CDR_Avg_Talk_Sec | source=CDR (agg) | Average duration of connected calls
- CDR_Max_Talk_Sec | source=CDR (agg) | Longest single call in seconds
- CDR_Distinct_Call_Days | source=CDR (agg) | Number of distinct days the lead was called
- CDR_Distinct_Agents | source=CDR (agg) | Distinct agent count (ids dropped)
- CDR_Callbacks_Scheduled | source=CDR Callback Time (agg) | Calls with a callback scheduled
- CDR_Inbound_Calls | source=CDR Call Direction (agg) | Inbound call count
- CDR_Top_Call_Status | source=CDR Call Status (agg) | Most frequent call status
- CDR_Top_Q_Type | source=CDR Q Type (agg) | Most frequent queue type
- CDR_First_Call_Delay_Days | source=CDR + CRM (derived) | Days from lead creation to first call
- CDR_Call_Span_Days | source=CDR (derived) | Days between first and last call
- MSG_Campaigns_Targeted | source=camp messaging (agg via customer key) | Distinct messaging campaigns that targeted this customer
- MSG_Sent | source=camp (agg) | Messages sent
- MSG_Delivered | source=camp (agg) | Messages delivered
- MSG_Read | source=camp (agg) | Messages read
- MSG_Clicked | source=camp (agg) | Link clicks
- MSG_Replied | source=camp (agg) | Replies
- MSG_Failed | source=camp (agg) | Failed + bounced messages
- MSG_Engaged | source=camp (derived) | 1 if any read/click/reply
- WEB_Tracked | source=WEB var5 (derived) | 1 if this lead had website activity (Apr-May window)
- Visits | source=WEB metrics | Web visits attributed to the lead id
- Page_Views | source=WEB metrics | Page views
- Total_Seconds_Spent | source=WEB metrics | Total seconds on site
- ev_* | source=WEB funnel events | Funnel event counts: lead creation/submission, calculator, quote, payment, proposal, OTP, brochure, expert-connect, scroll depth

- WEB_Step_Name | source=WEB var31 | Furthest journey step name recorded
- WEB_Step_Number | source=WEB var86 | Step number
- WEB_New_Vs_Repeat | source=WEB var3 | New vs repeat visitor
- WEB_Plan_Type | source=WEB var35 | Plan type explored
- WEB_Plan_Variant | source=WEB var105 | Plan variant explored
- WEB_Quoted_Price | source=WEB var90 | Top quoted price
- WEB_Coverage_Amount | source=WEB var91 | Top quoted coverage amount
- WEB_Payment_Frequency | source=WEB var97 | Payment frequency chosen
- WEB_Last_Touch_Channel | source=WEB lasttouchchannel | Last marketing touch channel
- WEB_UTM_Source | source=WEB campaign.utm-source | Web UTM source
- WEB_Device_Type | source=WEB mobiledevicetype | Device type used
- Label_Source_Disposition | source=CRM DISPOSITION | Raw disposition the label was derived from (drop before training to avoid leakage)
- Label_Source_Lead_Status | source=CRM LEAD_STATUS | Raw lead status (drop before training to avoid leakage)
- Label_Basis | source=derived | Which rule produced the label: conversion fields / disposition / lead_status
- LABEL_Customer_Validity | source=derived | TARGET: 0=junk/unreachable, 1=contactable not interested/eligible, 2=interested, 3=converted

4.3 Column To API Field Mapping

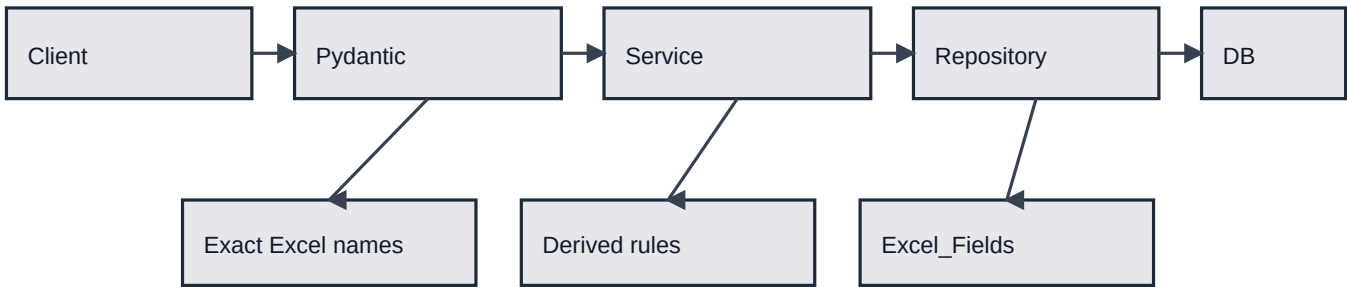
- Customer_ID -> customers.external_customer_id -> returned in /api/customers and /api/leads/{id}/details.customer
- CRM_Product_Code/CRM_Product_Name -> products.code/products.name -> returned in /api/products and details.campaign linkage
- CRM_UTM_Campaign/CRM_Data_Source_Platform -> campaigns.code/name -> returned in /api/campaigns
- CRM_Channel/CRM_Data_Medium/CRM_Source -> leads.source_channel/source_medium/current_handler
- Label_Source_Lead_Status/Label_Source_Disposition -> leads.status/recommended_action
- LABEL_Customer_Validity/CDR_Connect_Rate/Page_Views/MSG_Engaged -> leads.lead_score
- CRM_Department -> leads.current_section and lifecycle.assignment.to_section
- CDR_* metrics -> calls and engagement event payloads
- WEB_* metrics -> website_events fields and payload
- Label_Basis/Disposition -> task descriptions + followup notes + timeline details

5. How To Run In Detail

1. Create backend-local virtual environment and install requirements.
2. Configure .env with DATABASE_URL for PostgreSQL.
3. Run migrations: `./venv/bin/alembic upgrade head`
4. Preload real dataset: `./venv/bin/python scripts/preload_sample_data.py --reset`
5. Start API server: `./venv/bin/uvicorn app.main:app --host 0.0.0.0 --port 8000`
6. Open Swagger: /api/docs and OpenAPI: /api/openapi.json
7. Run smoke verification: `./venv/bin/python scripts/live_api_smoke.py`
8. Run tests: `./venv/bin/pytest -q`
9. Lifecycle-only test: `./venv/bin/pytest -q tests/test_lead_lifecycle.py`

API Flow Diagram:

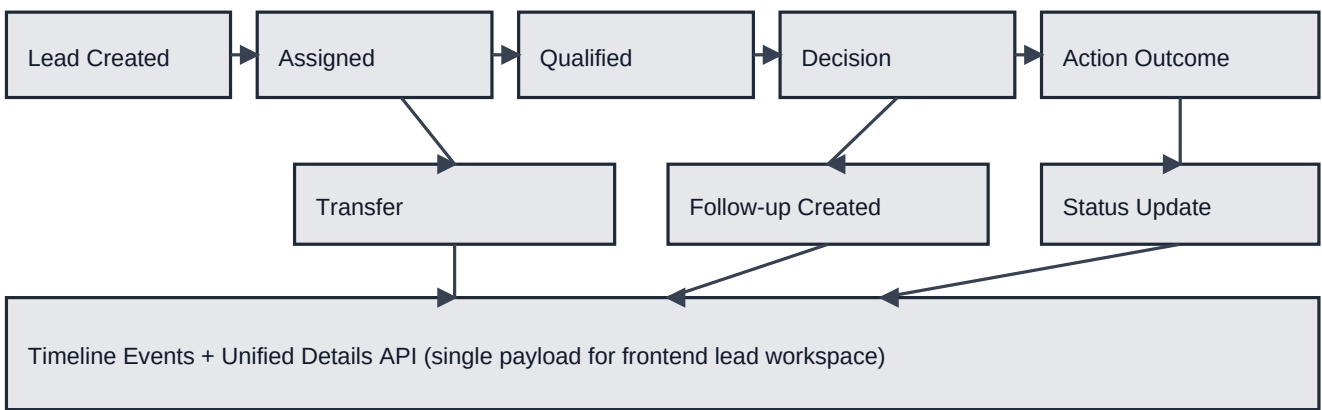
API Request And Response Flow



Response returns source fields plus clearly labeled system/AI outputs.

6. Feature Workflows And Handling

Lead Lifecycle Workflow



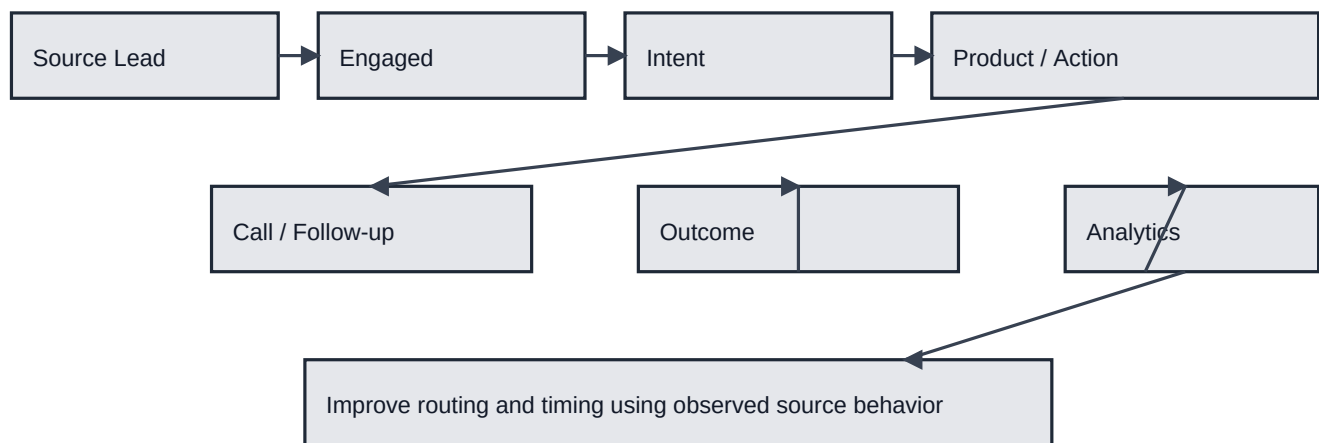
Outcomes can move lead stage, set recommendation, and trigger new operational tasks.

- 1. Lead intake: create lead -> initial timeline event -> assignment queue.
- 2. Assignment/transfer: updates current owner and section; previous assignment marked non-current.
- 3. AI scoring: prediction APIs persist score outputs per prediction type.
- 4. Decisioning: next action recommendation combines prediction signal and business logic.
- 5. Outcome handling: captures result, updates lead stage/status, optionally creates follow-up, and appends timeline events.
- 6. Reporting: analytics/reports endpoints query aggregated module-level data for operational monitoring.

Lifecycle Diagram (Mermaid):

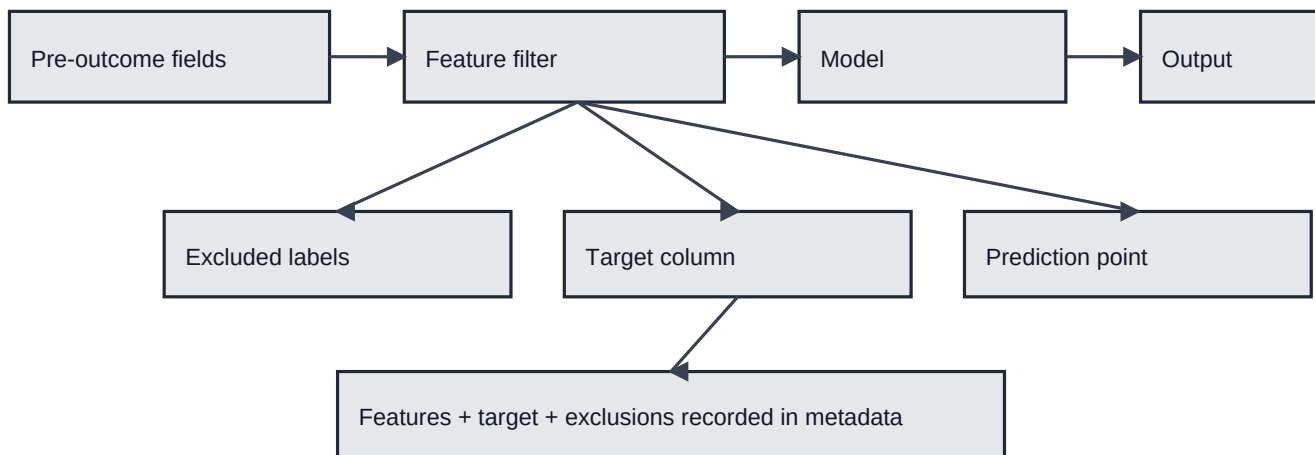
Sales Funnel Diagram:

Sales Funnel And Feedback



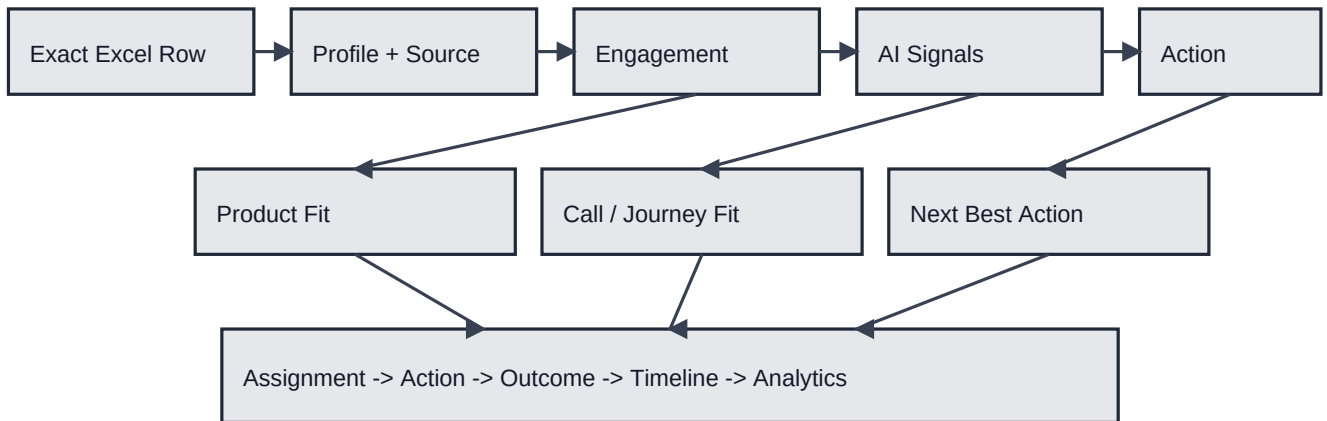
Prediction Guardrail Diagram:

Leakage-Safe Prediction Flow



7. Sales Optimization Playbook

Sales Optimization Flow



All decisions remain traceable to source columns or explicitly derived AI outputs.

The system maximizes productive sales activity by matching the next action to observed customer behavior, product context, contactability, and journey progress. It does not fabricate missing customer information or use outcome columns as prediction inputs.

7.1 Stage 1 - Source And Lead Intake

- Identify the lead using Customer_ID and associate source context from CRM_Channel, CRM_Department, CRM_Lead_Type, CRM_Lead_Source, CRM_Data_Medium, CRM_Source, and CRM_Data_Source_Platform.
- Preserve CRM_Product_Code, CRM_Product_Name, CRM_UTM_Source, CRM_UTM_Medium, and CRM_UTM_Campaign for attribution and product context.
- Use CRM_Lead_Create_Hour, CRM_Lead_Create_DayOfWeek, CRM_Lead_Create_Weekend, CRM_Lead_Create_Month, CRM_Days_Create_To_Update, CRM_Contact_Count, CRM_NonContact_Count, and CRM_No_Of_Attempts to understand timing and contact history.
- Do not use Label_Source_Disposition or Label_Source_Lead_Status as predictive inputs; they are observed labels/outcomes.

7.2 Stage 2 - Customer And Product Fit

- Build customer context from CRM_Gender, CRM_Age_Band, CRM_Income_Band, CRM_Occupation, CRM_Education, CRM_Tobacco_User, CRM_NonResident_Flag, and CRM_Existing_Plan_Flag.
- Match products using CRM_Product_Code, CRM_Product_Name, WEB_Plan_Type, WEB_Plan_Variant, WEB_Quoted_Price, WEB_Coverage_Amount, and WEB_Payment_Frequency.
- Product recommendation is marked unavailable when the source row does not contain enough product or website context.

7.3 Stage 3 - Engagement And Intent

- Measure messaging engagement from MSG_Campaigns_Targeted, MSG_Sent, MSG_Delivered, MSG_Read, MSG_Clicked, MSG_Replied, MSG_Failed, and MSG_Engaged.
- Measure call readiness from every CDR_field, including CDR_Total_Calls, CDR_Connected_Calls, CDR_Connect_Rate, CDR_Avg_Talk_Sec, CDR_Callbacks_Scheduled, CDR_Top_Call_Status, and CDR_Top_Q_Type.
- Measure digital intent from WEB_Tracked, Visits, Page_Views, Total_Seconds_Spent, WEB_Step_Name, WEB_Step_Number, WEB_New_Vs_Repeat, and WEB_Device_Type.

- Use ev_Lead_Creation through ev_Page_Scroll_95 to identify funnel progress, quote activity, form progress, OTP activity, brochure interest, expert connection, and payment journey signals.

7.4 Stage 4 - AI And Derived Lead Score

- Customer validity uses LABEL_Customer_Validity as the target only; it is never an input feature.
- Intent, engagement, segmentation, conversion, and lead score use only pre-outcome CRM, CDR, MSG, WEB, and ev_fields.
- CRM_Has_Application_No, CRM_Has_Payment_Flag, ev_Payment_Success, ev_Payment_Failure, Label_Source_Disposition, Label_Source_Lead_Status, and LABEL_Customer_Validity are excluded when they would leak the prediction target.
- Every prediction response records feature_columns, excluded_columns, target_column, prediction_point, and derived_output.

7.5 Stage 5 - Next Best Action

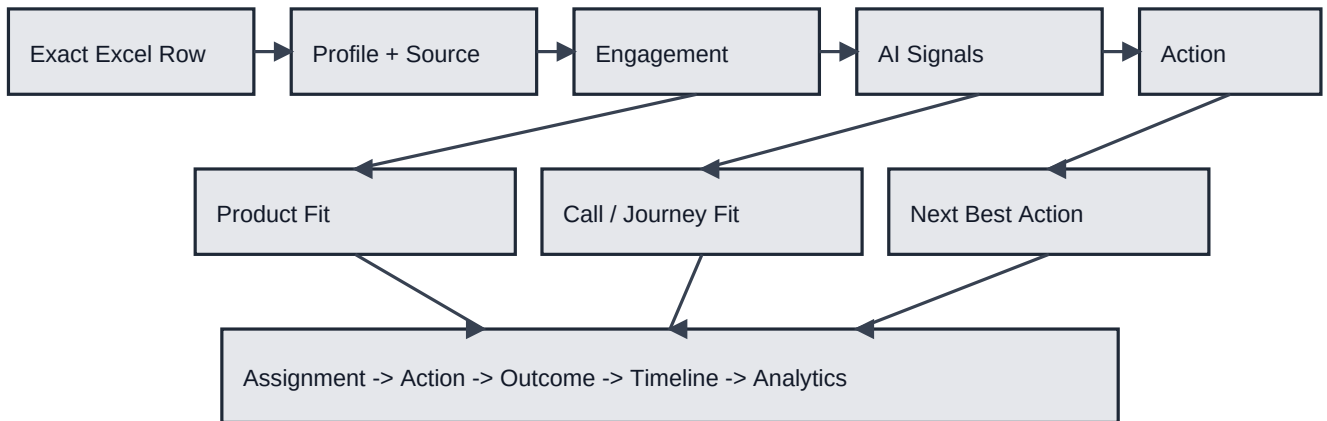
- High contactability and active journey progress should prioritize a timely call or expert handoff.
- Repeated visits, quote/proposal progress, or expert connection should prioritize product clarification and a focused follow-up.
- Read/click/reply or meaningful CDR activity should prioritize the channel with observed engagement rather than an assumed channel.
- Low activity should result in a measured nurture or information action, not an invented customer attribute or unsupported offer.
- The decision engine uses only Excel-derived signals and stored AI predictions; its recommendation is a derived system output.

7.6 Stage 6 - Assignment, Action, And Feedback

- Assign the lead to the operational section indicated by CRM_Department and preserve CRM source attribution.
- Record calls, tasks, follow-ups, and timeline events with the complete Excel_Fields snapshot attached.
- Capture outcome feedback only after an action occurs. Do not feed that outcome back into the same prediction request as an input.
- Use analytics and reports to compare source, product, journey, engagement, and call patterns for future campaign and staffing decisions.

Sales Optimization Diagram:

Sales Optimization Flow



All decisions remain traceable to source columns or explicitly derived AI outputs.

8. API Inventory And Response Contracts

Total API paths in OpenAPI: 62

POST /api/ai/analyze-lead

- Summary: Analyze Lead
- Tags: ai
- Request: body=context, customer_id, lead_id
- Responses: 200: customer_id, generated_at, lead_id, predictions | 422: detail

POST /api/ai/conversion

- Summary: Predict Conversion
- Tags: ai
- Request: body=context, customer_id, lead_id
- Responses: 200: details, label, model_name, prediction_type, rationale, score | 422: detail

GET /api/ai/customers/{customer_id}

- Summary: Get Customer Predictions
- Tags: ai
- Request: params=customer_id(path)
- Responses: 200: items[created_at], items[customer_id], items[details], items[id], items[label], items[lead_id], items[model_name], items[prediction_type], ... | 422: detail

POST /api/ai/intent

- Summary: Predict Intent
- Tags: ai
- Request: body=context, customer_id, lead_id
- Responses: 200: details, label, model_name, prediction_type, rationale, score | 422: detail

POST /api/ai/lead-score

- Summary: Predict Lead Score
- Tags: ai
- Request: body=context, customer_id, lead_id
- Responses: 200: details, label, model_name, prediction_type, rationale, score | 422: detail

GET /api/ai/leads/{lead_id}

- Summary: Get Lead Predictions
- Tags: ai
- Request: params=lead_id(path)
- Responses: 200: items[created_at], items[customer_id], items[details], items[id], items[label], items[lead_id], items[model_name], items[prediction_type], ... | 422: detail

POST /api/ai/next-best-action

- Summary: Predict Next Action
- Tags: ai
- Request: body=context, customer_id, lead_id
- Responses: 200: details, label, model_name, prediction_type, rationale, score | 422: detail

POST /api/ai/product-recommendation

- Summary: Predict Product
- Tags: ai
- Request: body=context, customer_id, lead_id
- Responses: 200: details, label, model_name, prediction_type, rationale, score | 422: detail

POST /api/ai/segment

- Summary: Predict Segment
- Tags: ai
- Request: body=context, customer_id, lead_id
- Responses: 200: details, label, model_name, prediction_type, rationale, score | 422: detail

POST /api/ai/validity

- Summary: Predict Validity
- Tags: ai
- Request: body=context, customer_id, lead_id
- Responses: 200: details, label, model_name, prediction_type, rationale, score | 422: detail

GET /api/analytics/channels

- Summary: Get Channels
- Tags: analytics
- Request: none
- Responses: 200: items

GET /api/analytics/funnel

- Summary: Get Funnel
- Tags: analytics
- Request: none
- Responses: 200: conversion_rate, converted_leads, lost_leads, new_leads, qualified_leads, total_leads

GET /api/analytics/overview

- Summary: Get Overview
- Tags: analytics
- Request: none
- Responses: 200: conversion_rate, converted_leads, open_tasks, pending_followups, total_calls, total_campaigns, total_customers, total_leads

GET /api/calls

- Summary: List Calls
- Tags: calls
- Request: params=skip(query), limit(query), lead_id(query), customer_id(query), status_filter(query)
- Responses: 200: items[CDR_Avg_Talk_Sec], items[CDR_Call_Direction], items[CDR_Top_Call_Status], items[Excel_Fields], items[created_at], items[customer_id], items[ended_at], items[id], ... | 422: detail

POST /api/calls

- Summary: Create Call
- Tags: calls
- Request: body=CDR_Call_Direction, CDR_Top_Call_Status, customer_id, lead_id, notes, phone_number, scheduled_at
- Responses: 201: CDR_Avg_Talk_Sec, CDR_Call_Direction, CDR_Top_Call_Status, Excel_Fields, created_at, customer_id, ended_at, id, ... | 422: detail

GET /api/calls/{call_id}

- Summary: Get Call
- Tags: calls
- Request: params=call_id(path)
- Responses: 200: CDR_Avg_Talk_Sec, CDR_Call_Direction, CDR_Top_Call_Status, Excel_Fields, created_at, customer_id, ended_at, id, ... | 422: detail

PATCH /api/calls/{call_id}

- Summary: Update Call
- Tags: calls
- Request: params=call_id(path) ; body=customer_id, direction, lead_id, notes, phone_number, scheduled_at, status
- Responses: 200: CDR_Avg_Talk_Sec, CDR_Call_Direction, CDR_Top_Call_Status, Excel_Fields, created_at, customer_id, ended_at, id, ... | 422: detail

DELETE /api/calls/{call_id}

- Summary: Delete Call

- Tags: calls
- Request: params=call_id(path)
- Responses: 204: Successful Response | 422: detail

POST /api/calls/{call_id}/end

- Summary: End Call
- Tags: calls
- Request: params=call_id(path) ; body=ended_at, notes
- Responses: 200: CDR_Avg_Talk_Sec, CDR_Call_Direction, CDR_Top_Call_Status, Excel_Fields, created_at, customer_id, ended_at, id, ... | 422: detail

POST /api/calls/{call_id}/start

- Summary: Start Call
- Tags: calls
- Request: params=call_id(path) ; body=started_at
- Responses: 200: CDR_Avg_Talk_Sec, CDR_Call_Direction, CDR_Top_Call_Status, Excel_Fields, created_at, customer_id, ended_at, id, ... | 422: detail

GET /api/campaigns

- Summary: List Campaigns
- Tags: campaigns
- Request: params=skip(query), limit(query)
- Responses: 200: items[CRM_UTM_Campaign], items[CRM_UTM_Source], items[Excel_Fields], items[code], items[created_at], items[end_date], items[id], items[start_date], ... | 422: detail

POST /api/campaigns

- Summary: Create Campaign
- Tags: campaigns
- Request: body=CRM_UTM_Campaign, CRM_UTM_Source, code, end_date, start_date, status
- Responses: 201: CRM_UTM_Campaign, CRM_UTM_Source, Excel_Fields, code, created_at, end_date, id, start_date, ... | 422: detail

GET /api/campaigns/{campaign_id}

- Summary: Get Campaign
- Tags: campaigns
- Request: params=campaign_id(path)
- Responses: 200: CRM_UTM_Campaign, CRM_UTM_Source, Excel_Fields, code, created_at, end_date, id, start_date, ... | 422: detail

PATCH /api/campaigns/{campaign_id}

- Summary: Update Campaign
- Tags: campaigns
- Request: params=campaign_id(path) ; body=channel, code, end_date, name, start_date, status
- Responses: 200: CRM_UTM_Campaign, CRM_UTM_Source, Excel_Fields, code, created_at, end_date, id, start_date, ... | 422: detail

DELETE /api/campaigns/{campaign_id}

- Summary: Delete Campaign
- Tags: campaigns
- Request: params=campaign_id(path)
- Responses: 204: Successful Response | 422: detail

GET /api/campaigns/{campaign_id}/leads

- Summary: List Campaign Leads
- Tags: campaigns
- Request: params=campaign_id(path), skip(query), limit(query)
- Responses: 200: items[CRM_Channel], items[CRM_Data_Medium], items[Excel_Fields], items[Label_Source_Lead_Status], items[campaign_id], items[created_at], items[current_handler], items[current_section], ... | 422: detail

GET /api/campaigns/{campaign_id}/performance

- Summary: Get Campaign Performance
- Tags: campaigns
- Request: params=campaign_id(path)
- Responses: 200: by_channel, by_metric, campaign_id, total_events | 422: detail

GET /api/customers

- Summary: List Customers
- Tags: customers
- Request: params=skip(query), limit(query)
- Responses: 200: items[CRM_Age_Band], items[CRM_Education], items[CRM_Existing_Plan_Flag], items[CRM_Gender], items[CRM_Income_Band], items[CRM_NonResident_Flag], items[CRM_Occupation], items[CRM_Tobacco_User], ... | 422: detail

POST /api/customers

- Summary: Create Customer
- Tags: customers
- Request: body=CRM_Age_Band, CRM_Education, CRM_Existing_Plan_Flag, CRM_Gender, CRM_Income_Band, CRM_NonResident_Flag, CRM_Occupation, CRM_Tobacco_User, Customer_ID
- Responses: 201: CRM_Age_Band, CRM_Education, CRM_Existing_Plan_Flag, CRM_Gender, CRM_Income_Band, CRM_NonResident_Flag, CRM_Occupation, CRM_Tobacco_User, ... | 422: detail

GET /api/customers/{customer_id}

- Summary: Get Customer
- Tags: customers
- Request: params=customer_id(path)
- Responses: 200: CRM_Age_Band, CRM_Education, CRM_Existing_Plan_Flag, CRM_Gender, CRM_Income_Band, CRM_NonResident_Flag, CRM_Occupation, CRM_Tobacco_User, ... | 422: detail

PATCH /api/customers/{customer_id}

- Summary: Update Customer
- Tags: customers
- Request: params=customer_id(path) ; body=CRM_Age_Band, CRM_Education, CRM_Existing_Plan_Flag, CRM_Gender, CRM_Income_Band, CRM_NonResident_Flag, CRM_Occupation, CRM_Tobacco_User, Customer_ID
- Responses: 200: CRM_Age_Band, CRM_Education, CRM_Existing_Plan_Flag, CRM_Gender, CRM_Income_Band, CRM_NonResident_Flag, CRM_Occupation, CRM_Tobacco_User, ... | 422: detail

DELETE /api/customers/{customer_id}

- Summary: Delete Customer
- Tags: customers
- Request: params=customer_id(path)
- Responses: 204: Successful Response | 422: detail

GET /api/dashboard

- Summary: Get Dashboard
- Tags: dashboard
- Request: none
- Responses: 200: conversion_rate, converted_leads, open_tasks, pending_followups, total_calls, total_campaigns, total_customers, total_leads

POST /api/data/export

- Summary: Export Data
- Tags: data
- Request: body=file_format, resource
- Responses: 201: file_format, file_path, job_id, resource, row_count, status | 422: detail

GET /api/data/history

- Summary: Get Data History
- Tags: data
- Request: params=skip(query), limit(query)
- Responses: 200: items[completed_at], items[created_at], items[dataset_name], items[details], items[error_message], items[id], items[invalid_row_count], items[row_count], ... | 422: detail

POST /api/data/import

- Summary: Import Data
- Tags: data
- Request: body=dataset_name, file_format, source_path
- Responses: 201: completed_at, created_at, dataset_name, details, error_message, id, invalid_row_count, row_count, ... | 422: detail

GET /api/data/import/{job_id}

- Summary: Get Import Job
- Tags: data
- Request: params=job_id(path)

- Responses: 200: completed_at, created_at, dataset_name, details, error_message, id, invalid_row_count, row_count, ... | 422: detail

GET /api/data/quality

- Summary: Get Data Quality
- Tags: data
- Request: none
- Responses: 200: completeness, generated_at, table_counts

GET /api/data/validation

- Summary: Get Data Validation
- Tags: data
- Request: none
- Responses: 200: generated_at, issues, total_issues

GET /api/decision/customers/{customer_id}

- Summary: Get Customer Decisions
- Tags: decision
- Request: params=customer_id(path)
- Responses: 200: items[confidence], items[created_at], items[customer_id], items[decision_type], items[id], items[inputs], items[lead_id], items[priority], ... | 422: detail

GET /api/decision/leads/{lead_id}

- Summary: Get Lead Decisions
- Tags: decision
- Request: params=lead_id(path)
- Responses: 200: items[confidence], items[created_at], items[customer_id], items[decision_type], items[id], items[inputs], items[lead_id], items[priority], ... | 422: detail

POST /api/decision/next-action

- Summary: Decide Next Action
- Tags: decision
- Request: body=customer_id, lead_id
- Responses: 200: confidence, decision_type, inputs, priority, rationale, recommended_action | 422: detail

GET /api/engagement/email

- Summary: List Email Events
- Tags: engagement
- Request: params=skip(query), limit(query), lead_id(query), customer_id(query), campaign_id(query)
- Responses: 200: items[CRM_Channel], items[Excel_Fields], items[MSG_Engaged], items[campaign_id], items[customer_id], items[event_payload], items[event_time], items[id], ... | 422: detail

GET /api/engagement/rcs

- Summary: List Rcs Events

- Tags: engagement
- Request: params=skip(query), limit(query), lead_id(query), customer_id(query), campaign_id(query)
- Responses: 200: items[CRM_Channel], items[Excel_Fields], items[MSG_Engaged], items[campaign_id], items[customer_id], items[event_payload], items[event_time], items[id], ... | 422: detail

GET /api/engagement/website

- Summary: List Website Events
- Tags: engagement
- Request: params=skip(query), limit(query), lead_id(query), customer_id(query), campaign_id(query)
- Responses: 200: items[CRM_Channel], items[Excel_Fields], items[MSG_Engaged], items[campaign_id], items[customer_id], items[event_payload], items[event_time], items[id], ... | 422: detail

GET /api/engagement/whatsapp

- Summary: List Whatsapp Events
- Tags: engagement
- Request: params=skip(query), limit(query), lead_id(query), customer_id(query), campaign_id(query)
- Responses: 200: items[CRM_Channel], items[Excel_Fields], items[MSG_Engaged], items[campaign_id], items[customer_id], items[event_payload], items[event_time], items[id], ... | 422: detail

GET /api/followups

- Summary: List Followups
- Tags: followups
- Request: params=skip(query), limit(query), task_id(query), lead_id(query), customer_id(query)
- Responses: 200: items[Excel_Fields], items[Label_Basis], items[channel], items[completed_at], items[created_at], items[customer_id], items[id], items[lead_id], ... | 422: detail

POST /api/followups

- Summary: Create Followup
- Tags: followups
- Request: body=Label_Basis, channel, customer_id, lead_id, scheduled_at, status, task_id
- Responses: 201: Excel_Fields, Label_Basis, channel, completed_at, created_at, customer_id, id, lead_id, ... | 422: detail

PATCH /api/followups/{followup_id}

- Summary: Update Followup
- Tags: followups
- Request: params=followup_id(path) ; body=channel, completed_at, customer_id, lead_id, notes, scheduled_at, status, task_id
- Responses: 200: Excel_Fields, Label_Basis, channel, completed_at, created_at, customer_id, id, lead_id, ... | 422: detail

GET /api/health

- Summary: Health Check
- Tags: health
- Request: none

- Responses: 200: database, status

GET /api/journey/leads/{lead_id}

- Summary: Get Lead Journey Summary
- Tags: journey
- Request: params=lead_id(path)
- Responses: 200: last_event_time, lead_id, total_events, unique_event_names | 422: detail

GET /api/journey/website

- Summary: List Website Events
- Tags: journey
- Request: params=skip(query), limit(query), lead_id(query), customer_id(query), device_type(query)
- Responses: 200: items[Excel_Fields], items[WEB_Device_Type], items[WEB_New_Vs_Repeat], items[WEB_Step_Name], items[WEB_Step_Number], items[customer_id], items[event_name], items[event_payload], ... | 422: detail

GET /api/leads

- Summary: List Leads
- Tags: leads
- Request: params=skip(query), limit(query), customer_id(query), campaign_id(query)
- Responses: 200: items[CRM_Channel], items[CRM_Data_Medium], items[Excel_Fields], items[Label_Source_Lead_Status], items[campaign_id], items[created_at], items[current_handler], items[current_section], ... | 422: detail

POST /api/leads

- Summary: Create Lead
- Tags: leads
- Request: body=CRM_Channel, CRM_Data_Medium, Label_Source_Lead_Status, campaign_id, current_handler, current_section, current_stage, customer_id, excel_fields, lead_score, ...
- Responses: 201: CRM_Channel, CRM_Data_Medium, Excel_Fields, Label_Source_Lead_Status, campaign_id, created_at, current_handler, current_section, ... | 422: detail

GET /api/leads/{lead_id}

- Summary: Get Lead
- Tags: leads
- Request: params=lead_id(path)
- Responses: 200: CRM_Channel, CRM_Data_Medium, Excel_Fields, Label_Source_Lead_Status, campaign_id, created_at, current_handler, current_section, ... | 422: detail

PATCH /api/leads/{lead_id}

- Summary: Update Lead
- Tags: leads
- Request: params=lead_id(path) ; body=campaign_id, current_handler, current_section, current_stage, customer_id, excel_fields, lead_score, priority, recommended_action, source_channel, ...

- Responses: 200: CRM_Channel, CRM_Data_Medium, Excel_Fields, Label_Source_Lead_Status, campaign_id, created_at, current_handler, current_section, ... | 422: detail

DELETE /api/leads/{lead_id}

- Summary: Delete Lead
- Tags: leads
- Request: params=lead_id(path)
- Responses: 204: Successful Response | 422: detail

POST /api/leads/{lead_id}/assign

- Summary: Assign Lead
- Tags: leads
- Request: params=lead_id(path) ; body=details, reason, related_decision_id, to_handler, to_section, trigger
- Responses: 200: Excel_Fields, assignment_type, created_at, customer_id, details, from_handler, from_section, id, ... | 422: detail

GET /api/leads/{lead_id}/calls

- Summary: Get Lead Calls
- Tags: leads
- Request: params=lead_id(path), skip(query), limit(query)
- Responses: 200: items[CDR_Avg_Talk_Sec], items[CDR_Call_Direction], items[CDR_Top_Call_Status], items[Excel_Fields], items[created_at], items[customer_id], items[ended_at], items[id], ... | 422: detail

GET /api/leads/{lead_id}/details

- Summary: Get Lead Details
- Tags: leads
- Request: params=lead_id(path)
- Responses: 200: ai, assignment, calls, campaign, customer, engagement, followups, journey, ... | 422: detail

GET /api/leads/{lead_id}/journey

- Summary: Get Lead Journey
- Tags: leads
- Request: params=lead_id(path), skip(query), limit(query)
- Responses: 200: items[Excel_Fields], items[WEB_Device_Type], items[WEB_New_Vs_Repeat], items[WEB_Step_Name], items[WEB_Step_Number], items[customer_id], items[event_name], items[event_payload], ... | 422: detail

GET /api/leads/{lead_id}/outcomes

- Summary: List Lead Outcomes
- Tags: leads
- Request: params=lead_id(path)
- Responses: 200: items[Excel_Fields], items[action_type], items[created_at], items[customer_id], items[details], items[followup_required], items[id], items[lead_id], ... | 422: detail

POST /api/leads/{lead_id}/outcomes

- Summary: Record Lead Outcome
- Tags: leads
- Request: params=lead_id(path) ; body=action_type, details, followup_required, next_action_hint, notes, outcome_code, outcome_label
- Responses: 201: Excel_Fields, action_type, created_at, customer_id, details, followup_required, id, lead_id, ... | 422: detail

GET /api/leads/{lead_id}/timeline

- Summary: Get Lead Timeline
- Tags: leads
- Request: params=lead_id(path)
- Responses: 200: items[Excel_Fields], items[created_at], items[customer_id], items[details], items[event_source], items[event_type], items[id], items[lead_id] | 422: detail

POST /api/leads/{lead_id}/transfer

- Summary: Transfer Lead
- Tags: leads
- Request: params=lead_id(path) ; body=details, reason, related_decision_id, to_handler, to_section, trigger
- Responses: 200: Excel_Fields, assignment_type, created_at, customer_id, details, from_handler, from_section, id, ... | 422: detail

GET /api/ml/models

- Summary: List Models
- Tags: ml
- Request: params=skip(query), limit(query)
- Responses: 200: items[created_at], items[latest_job_id], items[model_name], items[status], items[training_rows] | 422: detail

POST /api/ml/train

- Summary: Train Model
- Tags: ml
- Request: body=model_name
- Responses: 201: artifact_path, completed_at, created_at, error_message, id, metrics, model_name, status, ... | 422: detail

GET /api/ml/train/{job_id}

- Summary: Get Training Job
- Tags: ml
- Request: params=job_id(path)
- Responses: 200: artifact_path, completed_at, created_at, error_message, id, metrics, model_name, status, ... | 422: detail

GET /api/products

- Summary: List Products
- Tags: products

- Request: params=skip(query), limit(query)
- Responses: 200: items[CRM_Product_Code], items[CRM_Product_Name], items[Excel_Fields], items[created_at], items[id], items[updated_at] | 422: detail

POST /api/products

- Summary: Create Product
- Tags: products
- Request: body=CRM_Product_Code, CRM_Product_Name
- Responses: 201: CRM_Product_Code, CRM_Product_Name, Excel_Fields, created_at, id, updated_at | 422: detail

GET /api/products/{product_id}

- Summary: Get Product
- Tags: products
- Request: params=product_id(path)
- Responses: 200: CRM_Product_Code, CRM_Product_Name, Excel_Fields, created_at, id, updated_at | 422: detail

PATCH /api/products/{product_id}

- Summary: Update Product
- Tags: products
- Request: params=product_id(path) ; body=code, name
- Responses: 200: CRM_Product_Code, CRM_Product_Name, Excel_Fields, created_at, id, updated_at | 422: detail

DELETE /api/products/{product_id}

- Summary: Delete Product
- Tags: products
- Request: params=product_id(path)
- Responses: 204: Successful Response | 422: detail

GET /api/reports/campaign-performance

- Summary: Get Campaign Performance
- Tags: reports
- Request: none
- Responses: 200: items

GET /api/reports/pipeline

- Summary: Get Pipeline
- Tags: reports
- Request: none
- Responses: 200: converted_leads, lost_leads, new_leads, qualified_leads, total_leads

GET /api/reports/workload

- Summary: Get Workload
- Tags: reports

- Request: none
- Responses: 200: completed_followups, completed_tasks, in_progress_tasks, open_tasks, pending_followups, total_followups, total_tasks

GET /api/tasks

- Summary: List Tasks
- Tags: tasks
- Request: params=skip(query), limit(query), lead_id(query), customer_id(query), status_filter(query)
- Responses: 200: items[Excel_Fields], items[Label_Source_Disposition], items[created_at], items[customer_id], items[description], items[due_at], items[id], items[lead_id], ... | 422: detail

POST /api/tasks

- Summary: Create Task
- Tags: tasks
- Request: body=Label_Source_Disposition, customer_id, description, due_at, lead_id, priority, status
- Responses: 201: Excel_Fields, Label_Source_Disposition, created_at, customer_id, description, due_at, id, lead_id, ... | 422: detail

GET /api/tasks/{task_id}

- Summary: Get Task
- Tags: tasks
- Request: params=task_id(path)
- Responses: 200: Excel_Fields, Label_Source_Disposition, created_at, customer_id, description, due_at, id, lead_id, ... | 422: detail

PATCH /api/tasks/{task_id}

- Summary: Update Task
- Tags: tasks
- Request: params=task_id(path) ; body=customer_id, description, due_at, lead_id, priority, status, title
- Responses: 200: Excel_Fields, Label_Source_Disposition, created_at, customer_id, description, due_at, id, lead_id, ... | 422: detail

DELETE /api/tasks/{task_id}

- Summary: Delete Task
- Tags: tasks
- Request: params=task_id(path)
- Responses: 204: Successful Response | 422: detail

GET /health

- Summary: Health Check
- Tags: health
- Request: none
- Responses: 200: database, status

9. Lead Lifecycle Behavior

1. New lead creation writes lead_created event to timeline.
2. Assign/transfer updates lead ownership and section, and writes lifecycle events.
3. Outcome recording can update lead status/stage and auto-create follow-ups.
4. /leads/{id}/details returns a unified payload for frontend lead workspace rendering.

10. Validation Snapshot

- Test suite passing (including lifecycle tests).
- Live smoke APIs passing with real sample data preloaded.
- Lifecycle endpoints verified: timeline, details, assign, transfer, outcomes list/create.